

Thao Nguyen

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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Ph.D. Candidate in Computer Science

Aug. 2023 – Present

Advisor: Prof. Heng Ji

Hanoi University of Science and Technology (HUST)

Hanoi, Vietnam

B.S. in Electronics and Telecommunications Engineering

Aug. 2016 – Dec. 2020

Specialization: Biomedical Engineering

Advisor: Prof. Phuc Ngoc Pham

SELECTED PUBLICATIONS

9. iGC: Antibody Optimization via Reinforcement Learning over a Learned Somatic Hypermutation Prior

Thao Nguyen, Robert G. Alberstein, Ewa Nowara, Homa MohammadiPeyhani, Holly R. Steach, Sarah A. Robinson, Yuval Shechter, Donald W. Lee, Jason A. Vander Heiden, Franziska Seeger, Andrew Leaver-Fay, Richard Bonneau, Joshua Yao-Yu Lin
submitted to ICLR 2027

8. SpecOpt: Contact-Diff Reasoning for Agentic Molecule Optimization Toward Binding Specificity

Thao Nguyen, Heng Ji
preprint arXiv:2609.21165, 2026

7. EnSol: An Environment-Aware Graph Neural Network for Molecular Solubility Prediction

Thao Nguyen, Saman Shafaei, Zhengyi Zhang, Huimin Zhao, Heng Ji
preprint arXiv:2609.21151, 2026

6. Fraglingo: Molecular Design via Attachment-Aware Autoregressive Fragment Generation

Thao Nguyen, Jeonghwan Kim, Zhenhailong Wang, Heng Ji
preprint arXiv:2609.13519, 2026

5. MolLingo: Molecule-Native Representations for LLM-Powered Scientific Agents

Thao Nguyen, Heng Ji
preprint arXiv:2605.27853, 2026

4. FARM: Enhancing Molecular Representations with Functional Group Awareness

Thao Nguyen, Kuan-Hao Huang, Ge Liu, Martin D. Burke, Ying Diao, Heng Ji
Transactions on Machine Learning Research, 2026

3. ProteinZero: Self-Improving Protein Generation via Online Reinforcement Learning

Ziwen Wang, Jiajun Fan, Ruihan Guo, **Thao Nguyen**, Heng Ji, Ge Liu
Transactions on Machine Learning Research, 2026

2. mCLM: A Function-Infused and Synthesis-Friendly Modular Chemical Language Model

Carl Edwards*, Chi Han*, Gawon Lee, **Thao Nguyen**, Bowen Jin, Chetan Kumar Prasad, Sara Szymkuć, Bartosz A Grzybowski, Ying Diao, Jiawei Han, Ge Liu, Hao Peng, Martin D Burke, Heng Ji
Proceedings of the Fourteenth International Conference on Learning Representations (ICLR) (Oral Presentation), 2026

1. Variational Supervised Contrastive Learning

Ziwen Wang, Jiajun Fan, **Thao Nguyen**, Heng Ji, Ge Liu
Advances in Neural Information Processing Systems (NeurIPS), 2025

EXPERIENCE

Apodex

Research Intern

Remote

Sep. 2026 – Present

- Working on an agentic AI system for drug discovery and drug repurposing, integrating LLMs, molecular reasoning, and optimization workflows for therapeutic prioritization.

Prescient Design, Genentech / Roche

Research Intern

South San Francisco, CA

Summer 2026

- Worked on AI-driven methods for antibody design and optimization, including learning-based sequence and structure-informed models for therapeutic discovery.

BLENDER Lab, MMLI, CABBI, IIDAI

Research Assistant

University of Illinois Urbana-Champaign

2023 – Present

- Developing agentic AI systems for drug discovery, molecular optimization, and therapeutic prioritization across small molecules, natural products, and protein design.
- Building and evaluating foundation models for molecular representation, structure-aware generation, and property prediction in scientific discovery workflows.
- Working on closed-loop discovery pipelines that integrate LLMs, generative models, prediction oracles, and experimental feedback for real-world drug design tasks.

VinUni-Illinois Smart Health Center - VISHC

Research Assistant

VinUniversity

2022 – 2023

- Collected a real-world 3-lead ECG dataset and developed an algorithm to digitize the signals.
- Built high-accuracy classification models for 12 cardiovascular diseases using 3-lead ECG signals and integrated them into a mobile application.
- Performed biomedical signal processing for sleep apnea detection and sleep stage classification.

Department of Medical Image Processing

Intern

VinBigdata Institute

2020 – 2021

- Collect a dataset of 8,000 chest X-ray images and their corresponding medical reports from electronic health records (EHRs) in a provincial general hospital.
- Develop a pipeline to automatically label medical images using data sourced from EHRs.
- Chest X-ray images processing
- Build a robust system capable of accurately classifying Chest X-ray images according to specific regions of pathology identified in the images.

EEG and Rehabilitation Laboratory

Undergraduate Research Assistant

Hanoi University of Science and Technology

2017 – 2020

- Biomedical signal and image processing.

TEACHING ASSISTANT EXPERIENCE

Natural Language Processing @VinUniversity

Algorithms and Data Structures for Data Science @UIUC

PROFESSIONAL SERVICES

Session Chair

- CIKM 2024

Reviewer

- ICLR 2027
- ICLR 2026
- EMNLP 2025
- NeurIPS 2025

- ICML 2025
- ACL Rolling Review 2025
- ICLR 2025
- AI4Research Workshop @ AAAI 2025 - Best Reviewer Award
- Language + Molecules Workshop @ ACL 2024

REFERENCES

Prof. Heng Ji

Professor

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University of Illinois Urbana-Champaign

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Prof. Huimin Zhao

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Department of Chemical & Biomolecular Engineering

University of Illinois Urbana-Champaign

Email: *zhao5@illinois.edu*